

2017-Ph-H1-Q8

Section: Our Dynamic Universe

Topic: Gravitation

A satellite is raised to a higher orbit. What happens to its gravitational potential energy?

 **Key Concept:**

Gravitational potential energy (GPE) in space is:

$$E_p = -\frac{GMm}{r}$$

As r increases, the value becomes **less negative** → so GPE **increases**.

 **Final Answer: C — GPE increases**

 **Revision Tips:**

- GPE in space is **always negative**
 - Moving further from a planet = **increased GPE** (towards zero)
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